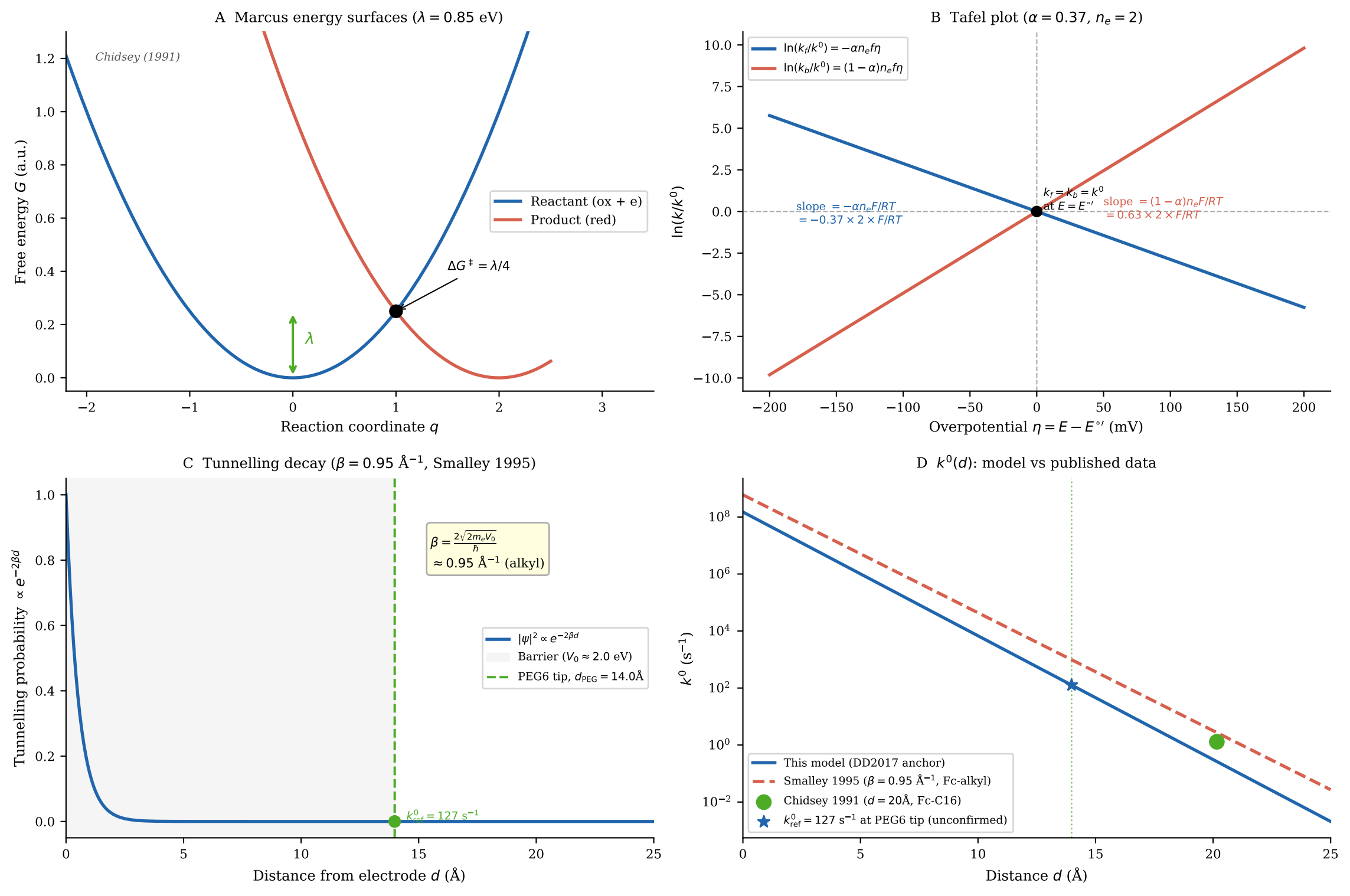

A Marcus energy surfaces ($\lambda = 0.85$ eV)
Chidsey (1991)
Free energy G (a.u.)
Reaction coordinate q
Reactant (ox + e)
Product (red)
$\Delta G^{\ddagger} = \lambda/4$
λ
B Tafel plot ($\alpha = 0.37$, $n_e = 2$)
$\ln(k_f/k^0) = -\alpha n_e f\eta$
$\ln(k_b/k^0) = (1-\alpha) n_e f\eta$
$\ln(k/k^0)$
slope $= -\alpha n_e F/RT$
$= -0.37 \times 2 \times F/RT$
$k_f = k_b = k^0$
at $E = E^{\circ\prime}$
slope $= (1-\alpha) n_e F/RT$
$= 0.63 \times 2 \times F/RT$
Overpotential $\eta = E - E^{\circ\prime}$ (mV)
C Tunnelling decay ($\beta = 0.95$ Å^{-1}, Smalley 1995)
Tunnelling probability $\propto e^{-2\beta d}$
$\beta = \frac{2\sqrt{2 m_e V_0}}{\hbar}$
≈ 0.95 Å^{-1} (alkyl)
$|\psi|^2 \propto e^{-2\beta d}$
Barrier ($V_0 \approx 2.0$ eV)
PEG6 tip, $d_{PEG} = 14.0$Å
$k^0_{ref} = 127$ s^{-1}
Distance from electrode d (Å)
D $k^0(d)$: model vs published data
k^0 (s^{-1})
This model (DD2017 anchor)
Smalley 1995 ($\beta = 0.95$ Å^{-1}, Fc-alkyl)
Chidsey 1991 ($d = 20$Å, Fc-C16)
$k^0_{ref} = 127$ s^{-1} at PEG6 tip (unconfirmed)
Distance d (Å)